Week 6

$$A \times B = (A_x i + A_y j + A_z k) \times (B_x i + B_y j + B_z k)$$

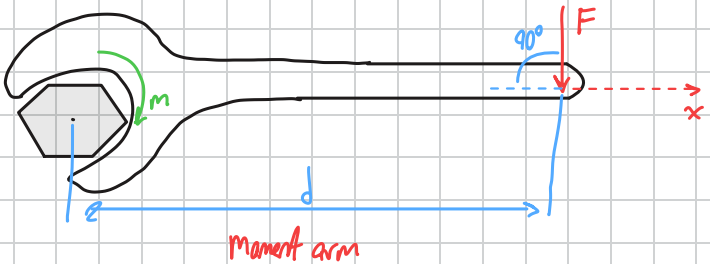
$$\vec{C} = (A_y B_z - A_z B_y) i - (A_x B_z - A_z B_x) j + (A_x B_y - A_y B_x) k$$

$$A \times B = \det \begin{bmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{bmatrix} = (A_y B_z - A_z B_y) i - (A_x B_z - A_z B_x) j + (A_x B_y - A_y B_x) k$$

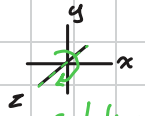
Moments Leeture 8

huge focus
once you unterstant youre good

=> Computing the moment of a force about a point
=> Computing the moment of a force about a specific axis



M is on the Z axis
So it stays on plane



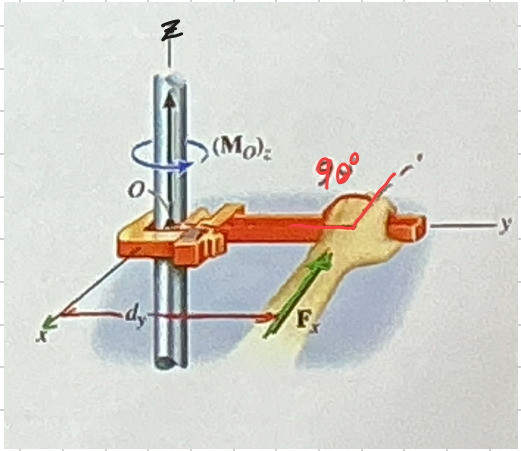
rotating around Z

2d → $M = F \cdot d$

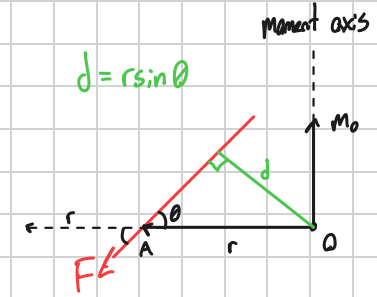
d is easly Calculated

this is why a torque bar works very well beause d is far greater requiring less F to turn

3D moment of force



Use Cross product to Calculate moments in 3D



$d = r \sin\theta$

$M_O = \vec{r} \times \vec{F}$

$= |\vec{r}||\vec{F}| \sin\theta$

$M_O = |\vec{F}| \cdot d$

$M_O = r \times F$

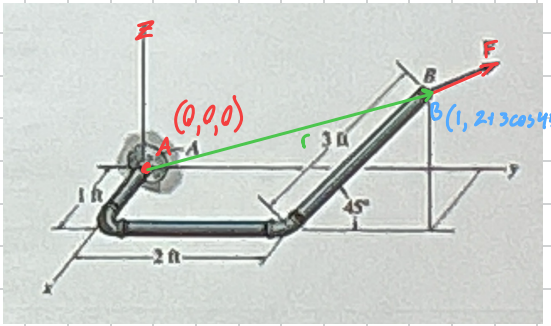
r is a position vector from O to any point on the line of action of F (use one you know)

$\vec{M}$ is a vector about spesific point

the resultant moment of forces

$$M_{RO} = \sum (r \times F)$$

Ex



$F = \{50i + 60j + 30k\}$

① M at A

$\vec{M}_A = \vec{r}_{AB} \times \vec{F}$

1) $\vec{r}_{AB} =$

2) $\vec{M}_B \longrightarrow$ from ①

α, β, γ ?

$\vec{u}_{\vec{M}_A} = \frac{\vec{M}_A}{|\vec{M}_A|} = \{ \cdots$

$\vec{r}_{AB} = \{(1-0)i + (4.121)j + (2.121)k\}$

$\vec{M}_B = \vec{r}_{AB} \times \vec{F}$

$$\begin{vmatrix} i & j & k \\ 1 & 4.121 & 2.121 \\ 50 & 60 & 30 \end{vmatrix}$$

$\vec{M}_B = [(2.121)(30) - (2.121)(60)]i -$ ~~~

$= \{-3.64i + 76.07j - 146.07k\}$ lb·ft

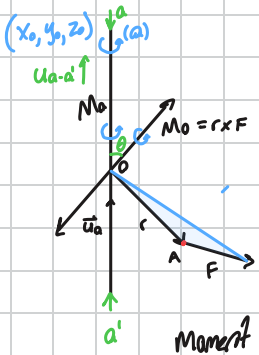


$M = F \cdot d$

$\vec{M_O} = \vec{r_{OA}} \times \vec{F}$

Dot product

Rigid body



Magnitude of moment about a a'

$M_a = m_O \cdot u_a$

Scalar

Moment @ point $M_O = r \times F$

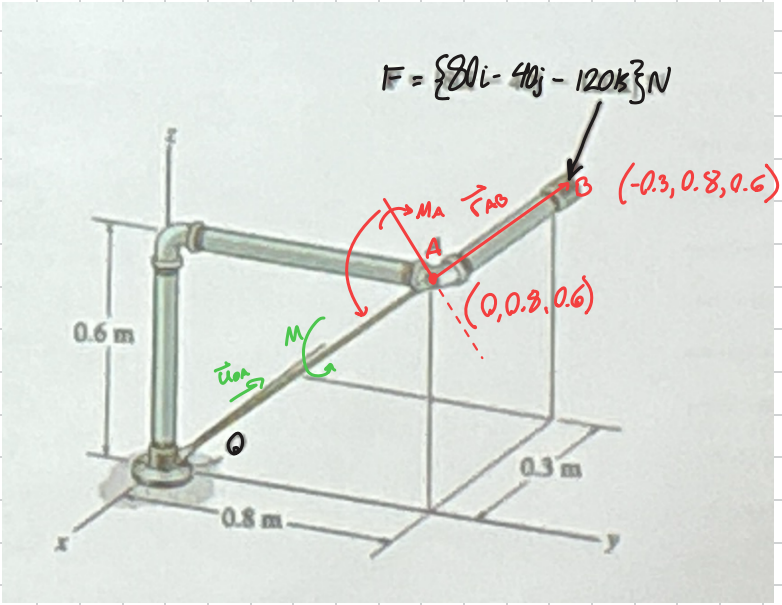
Express as cartesian vector

$$M_a = u_a \cdot (r \times F) = \begin{vmatrix} u_{ax} & u_{ay} & u_{az} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

$\vec{u}_{a-a'}$, $\vec{r}$, $\vec{F}$

if you get positive direction is correct

$\vec{r}_{OB}$ can also be used



1) $\vec{M}_A = \vec{r}_{AB} \times \vec{F}$

Point

2) $M_{OA} = \vec{u}_{OA} \cdot (\vec{r}_{AB} \times \vec{F})$

1) $\vec{r}_{AB} = \{-0.3\hat{i} + 0\hat{j} + 0\hat{k}\}$

2) $\vec{F} = \{80i - 40j - 120\hat{k}\}$

3) $\vec{u}_{OA} = \dfrac{\vec{r}_{OA}}{|\vec{r}_{OA}|}$

$\vec{r}_{OA} = \{0\hat{i} + 0.8\hat{j} + 0.6\hat{j}\}$

$|\vec{r}_{OA}| = \sqrt{0^2 + 0.8^2 + 0.6^2}$

$= 1$

$\vec{u}_{OA} = \{0i + 0.8j + 0.6k\}$

$$M_{OA} = \begin{vmatrix} 0 & 0.8 & 0.6 \\ -0.3 & 0 & 0 \\ 80 & -40 & -120 \end{vmatrix} \begin{matrix} \vec{u}_{OA} \\ \vec{r}_{AB} \\ \vec{F} \end{matrix}$$

$= \{(0)(-120) - (0)(-40)\} +$

$\{(0.3)(-120) - (0)(8)\} +$

$\{(-0.3)(-40) - (0)(80)\} = -21.6 \text{ N·m}$

Our Assumption was wrong. it's the opposite direction